01 / PROBLEM AND CONTRIBUTION

MOS characterization & resistive-load amplifier

EEE 433/591 · Analog IC Design · Lab 1 | Fall 2025

RESULT AT A GLANCE

Connected transistor operating-point analysis to amplifier behavior, reporting 21.38 dB transient voltage gain for a 10 kHz small-signal test.

Engineering question

How do MOS operating conditions and load resistance shape a common-source amplifier?

Situation

A useful amplifier bias point must balance gain and output headroom. This lab starts with a transistor characterization and carries those observations into a resistive-load common-source stage.

Task

Characterize the NMOS, establish the amplifier bias, and compare DC, AC, transient, and load-sweep behavior.

My contribution

Prepared transistor and amplifier schematics, operating-point comparisons, transfer curves, and transient/AC plots in the submitted Cadence lab report.

Tools and concepts

Cadence Virtuoso / MOS characterization / Common-source amplifiers / DC operating-point analysis / Transient analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Compared simulated NMOS operating-point quantities with hand analysis. The report lists 101.1 µA drain current, 507.07 µS transconductance, and 64.23 kΩ output resistance for the characterized device.

02 / Implementation

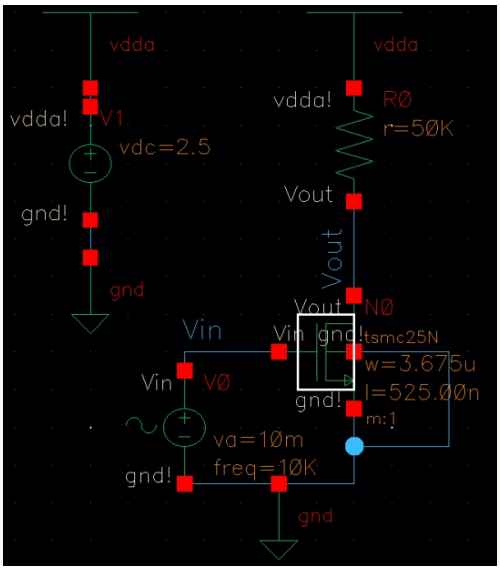
Evaluated the resistive-load stage with DC transfer curves and a 10 kHz sinusoidal input. Used the reported 20 mVpp input and 234.5 mVpp output to calculate a gain magnitude of 11.725 V/V, or 21.38 dB.

03 / Implementation

Examined input-amplitude and load-resistance sweeps to relate compression and clipping to bias and to describe the gain-versus-bandwidth tradeoff.

From device to circuit

The transistor operating point describes local small-signal behavior. The amplifier then adds a load resistor and an output node; the gain calculation must use the input and output from the amplifier experiment.



Resistive-load common-source schematic

Source report p. 6: NMOS stage and load used in the circuit study.

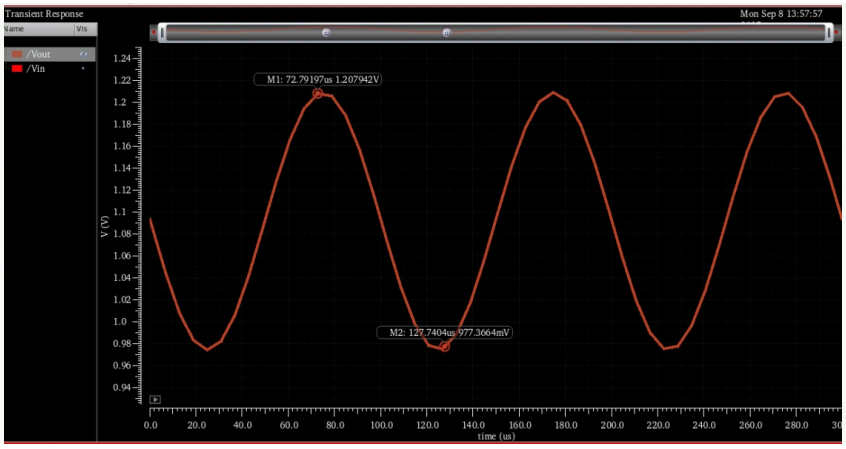
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

Transient voltage gain	21.38 dB
Input / output swing	20 / 234.5 mVpp
NMOS drain current	101.1 µA

Reported result

The report gives 21.38 dB transient gain for the stated small-signal test. The device comparison lists 101.1 µA simulated drain current against 100 µA in the hand calculation. These are separate device and amplifier observations.



Output transient waveform

Source report p. 9: output waveform for the documented small-signal experiment.

Scope and limitations

The gain is a transient amplitude-ratio result, not an independently verified AC bandwidth specification. The report contains inconsistent phase and signal-range wording, so no phase or maximum linear-swing claim is promoted.

Possible next steps - proposed work

- Reconcile the AC phase wording and annotate a consistent gain/bandwidth testbench.
- Record distortion against input amplitude before specifying a maximum linear signal range.

Source material: EEE 433/591 Lab 1 submitted report, physical PDF pp. 3-4, 6, 9-10, 13-14.